THE UNIVERSITY OF DANANG
UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF ELECTRONICS AND ARTIFICIAL INTELLIGENCE

Design of a Multi-Mode PWM Control and Monitoring System via UART on the KIWI 1P5 FPGA Board
Programming and Implementation of a Digital Clock on the SN32F407_EVK Board

Team name: Chilli

Team members:

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- Pham Huynh Duc
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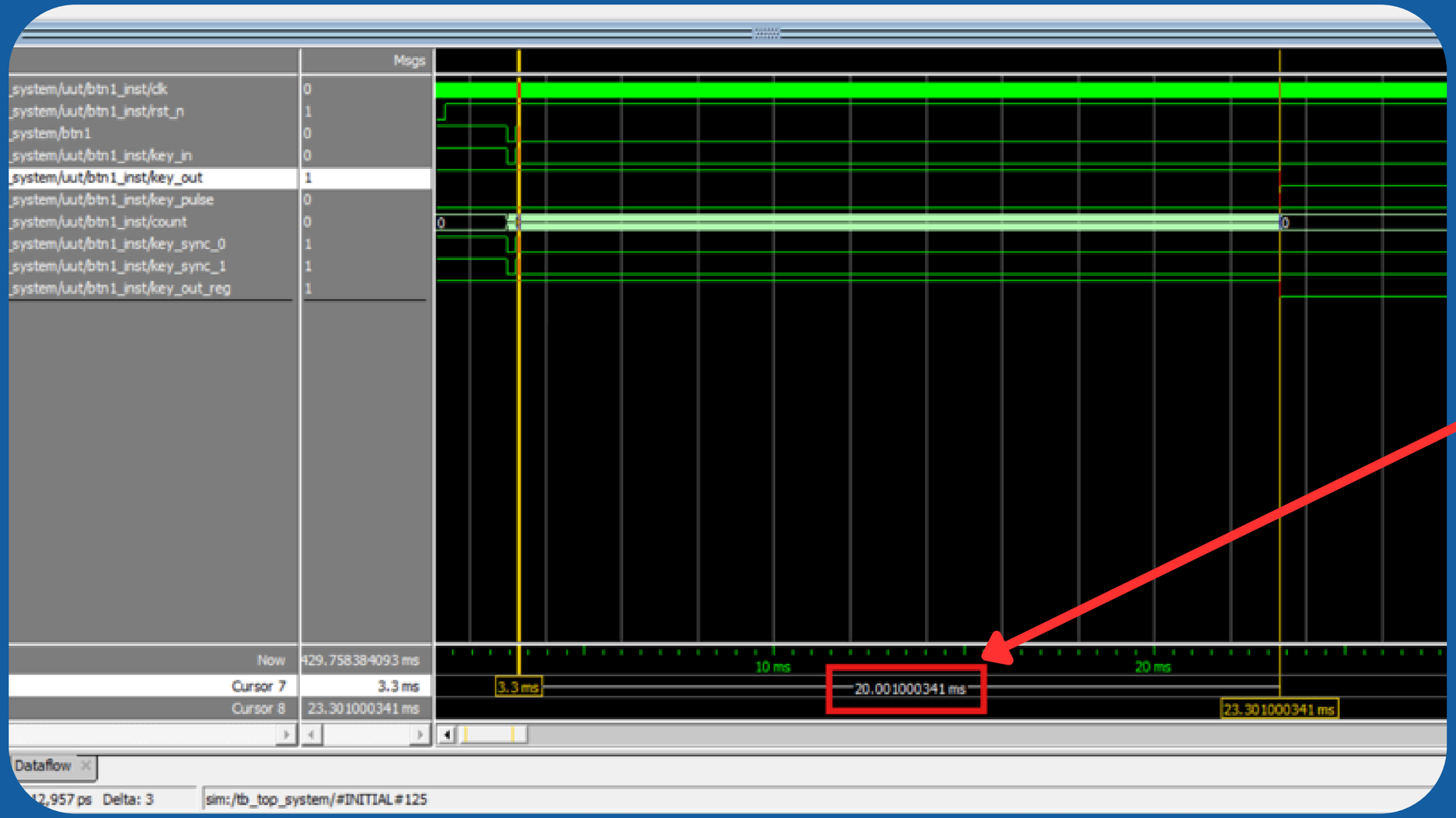
FPGA

DEBOUNCE MODULE SIMULATION



20 ms
DEBOUNCE INTERVAL
The input is accepted only after the 20 ms stability period has elapsed.

PWM LED SIMULATION

25% DUTY CYCLE VERIFICATION

Duty Cycle = (12,535 / 50,143) × 100%

≈ 24.9985% ≈ 25%

The result confirms that the PWM module correctly generates the required 25% duty cycle in LOW mode.

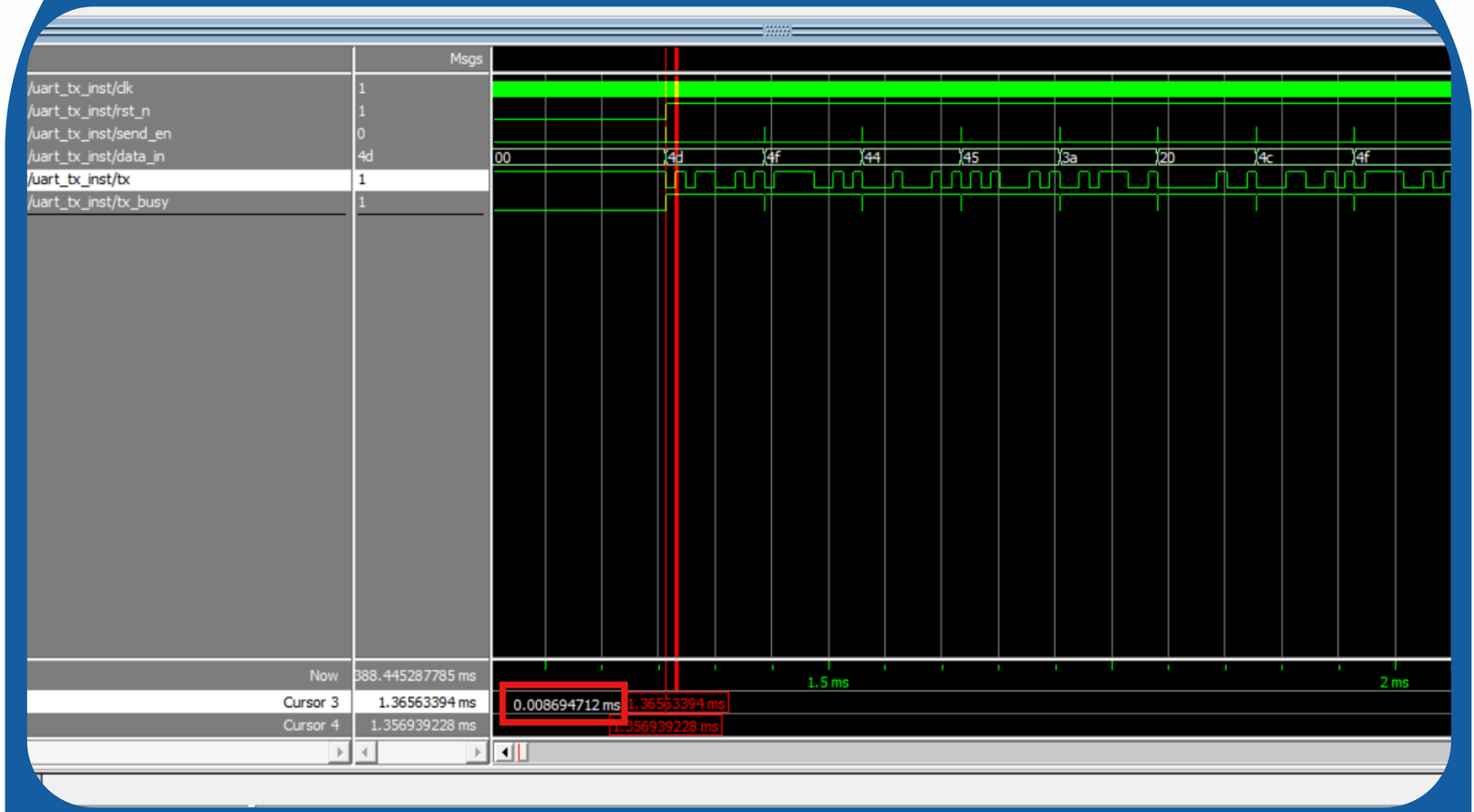
PWM LED SIMULATION

100% DUTY CYCLE VERIFICATION
Duty Cycle = 50,143
The result confirms that the PWM module correctly generates the required 100% duty cycle in HIGH mode.

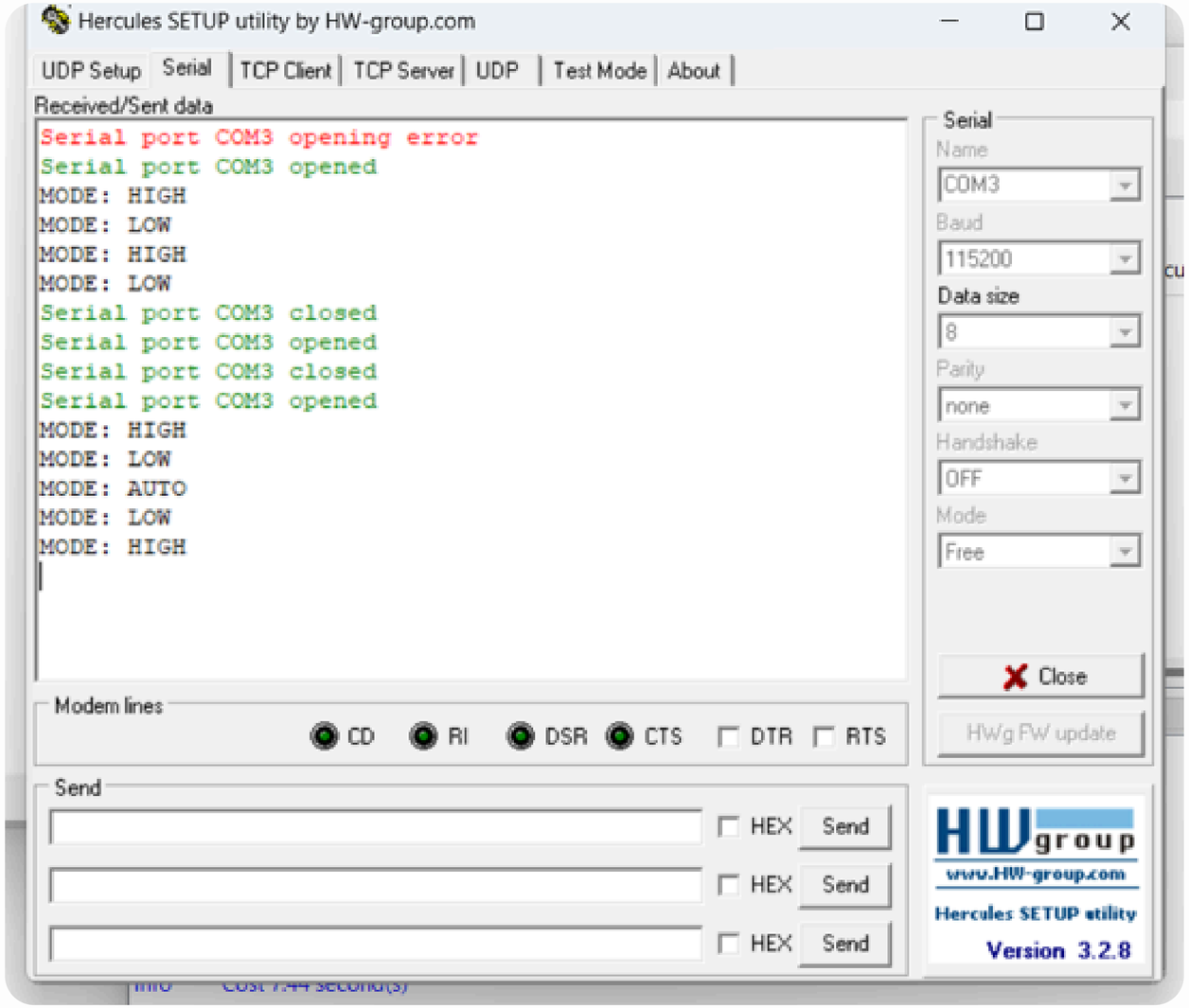
UART TX Simulation

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nullam laoreet risus fringilla, egestas elit a, consequat augue. Phasellus sollicitudin felis mi, quis egestas ex ornare sed.

Praesent vel felis quis mi pulvinar sagittis. Pellentesque viverra ipsum ante, in congue ante eleifend eget. Aenean tortor tellus, efficitur ac rhoncus et, eleifend sit amet mi. Sed ut lectus ac nibh molestie rhoncus.



UART Serial Communication



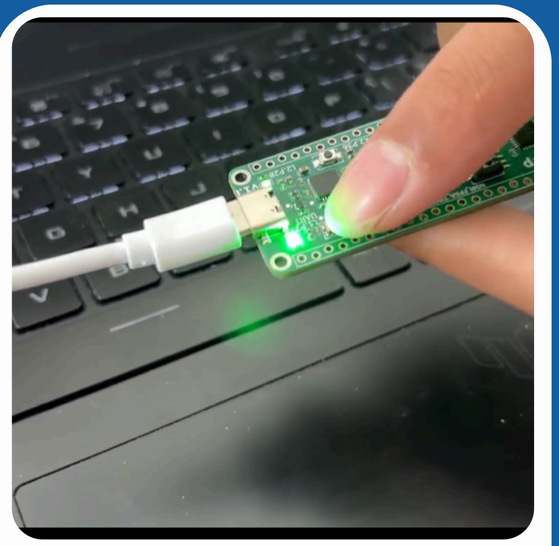
Hercules Serial Terminal:

- COM Port: COM3
- Baud Rate: 115200 bps
- Data Size: 8 bits
- Parity: None
- Stop Bits: 1
- UART format: 8-N-1

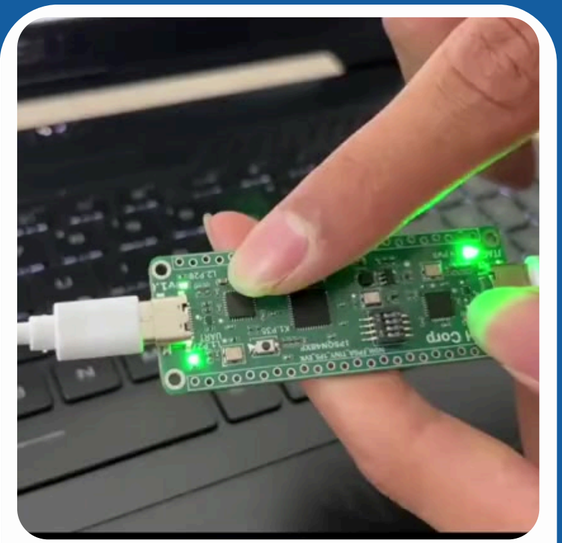
UART Serial Communication

- **LOW: Fixed 25% duty cycle for low LED brightness.**
- **HIGH: Fixed 100% duty cycle for maximum brightness.**
- **AUTO: Breathing effect with duty cycle changing from 0% → 100% → 0%.**

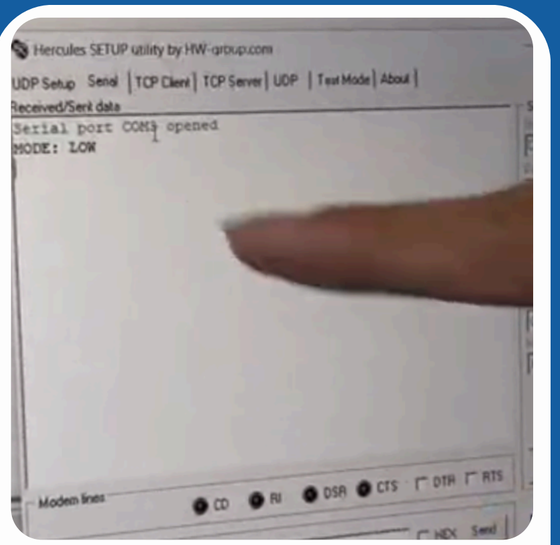
mode LOW in board



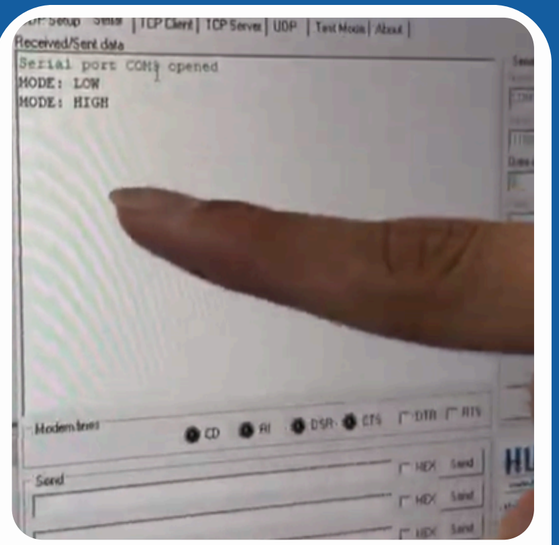
mode HIGH in board



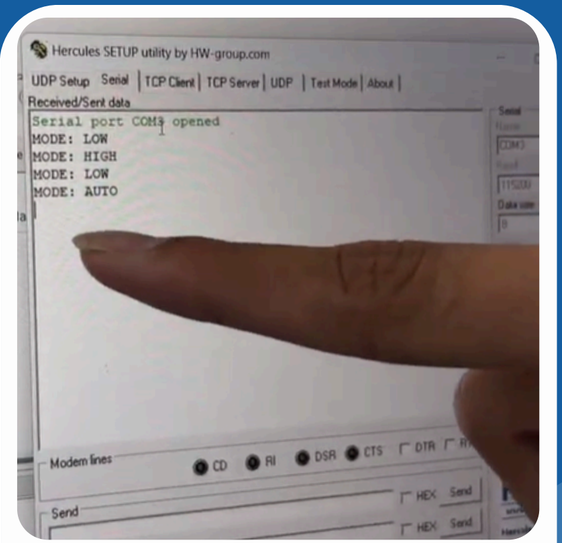
mode AUTO in board



mode LOW in Hercules



mode HIGH in Hercules



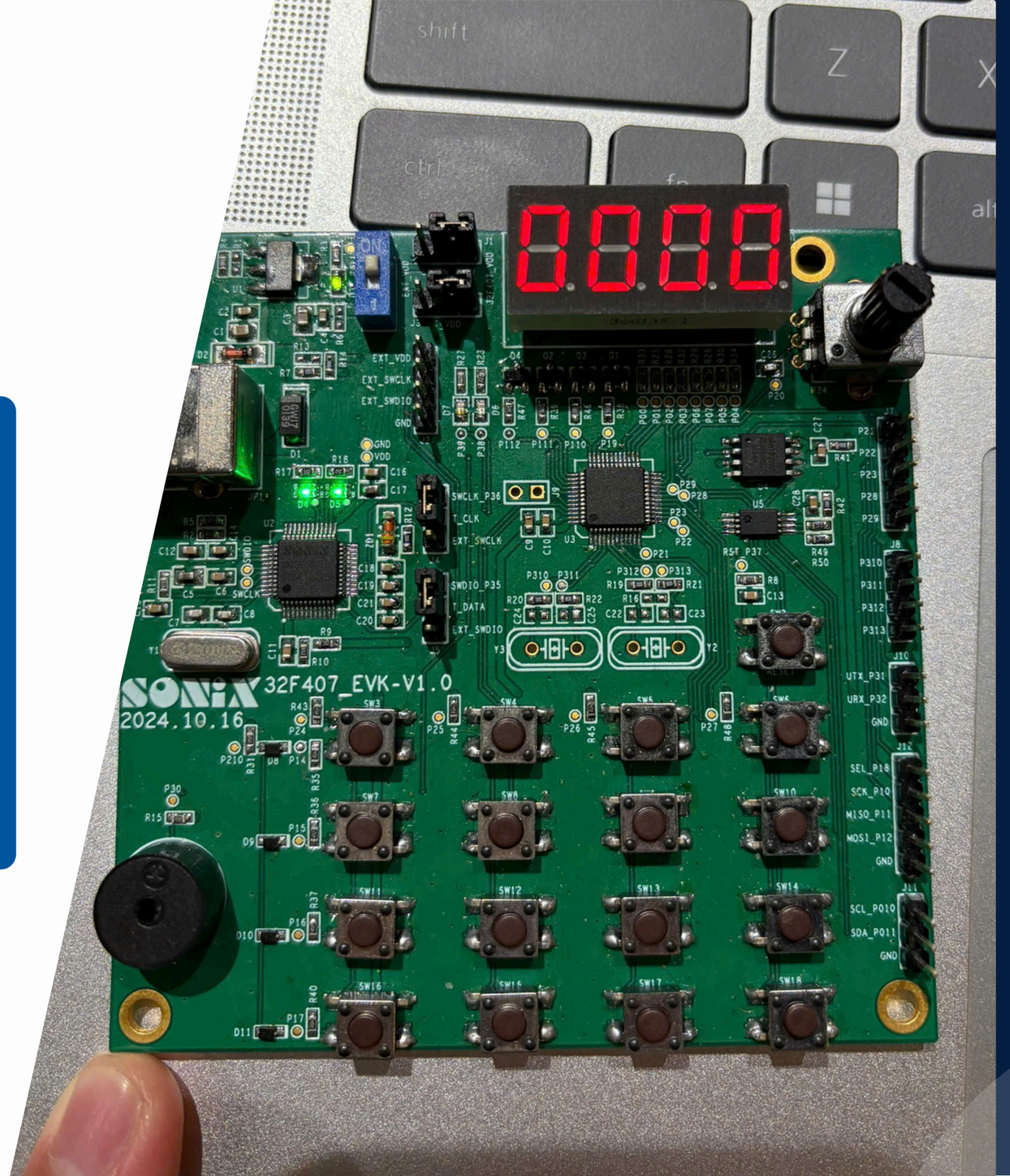
mode AUTO in Hercules

MCU

Requirement 1:

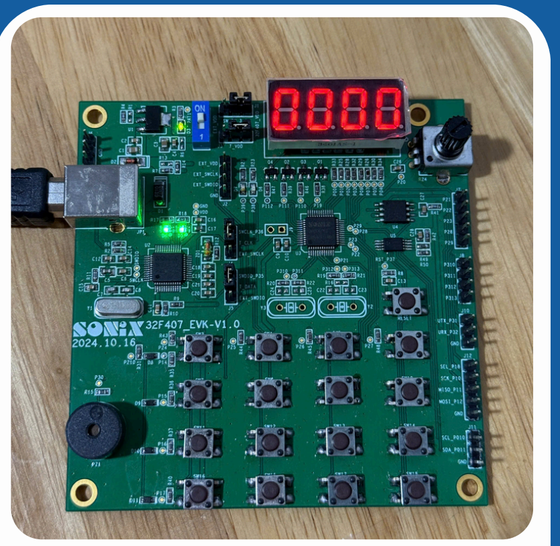
Problem 01

- Default display: Use four 7-segment LEDs to display hours and minutes in the HH.MM format.
- Startup: Upon power-up, the circuit operates in normal mode, with hours and minutes reset to 00:00.

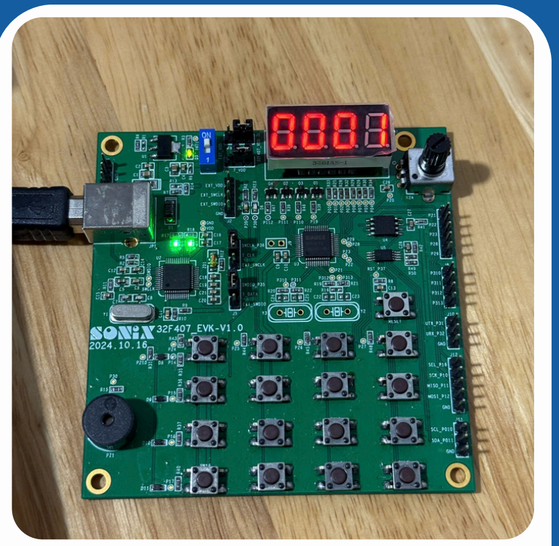


Problem 02

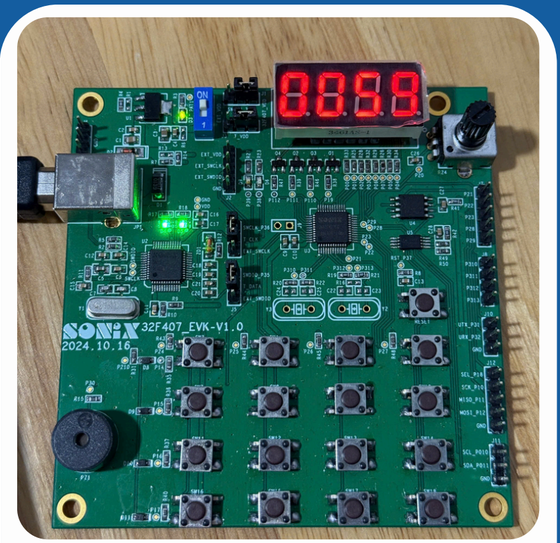
·Timer: The minute count increases by 1 every minute. Upon reaching the 60th minute, the minutes reset to 0 and the hour increases by 1. The hour is displayed in the range of 00 to 23.



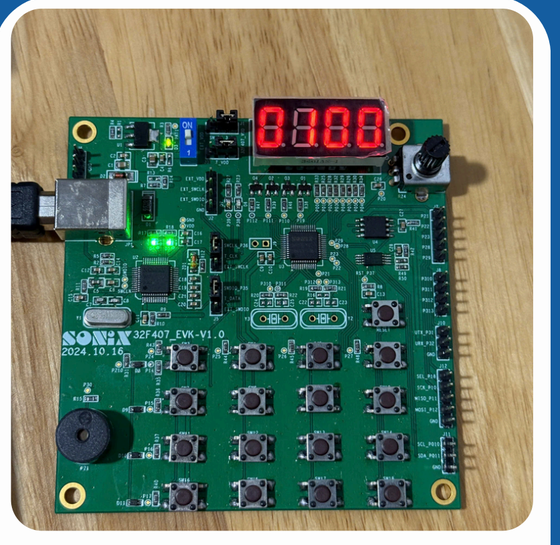
Start



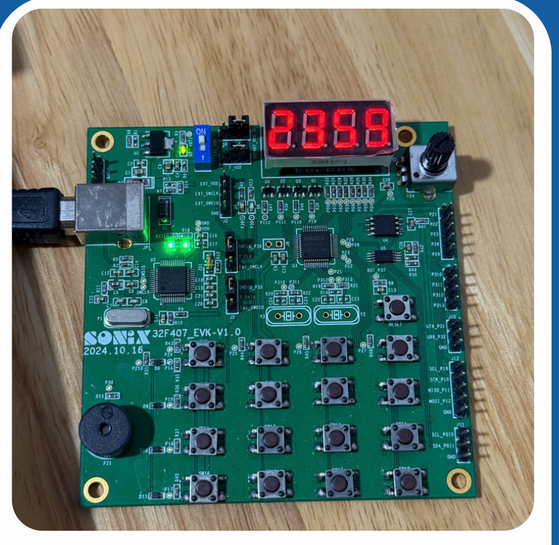
1 Minute



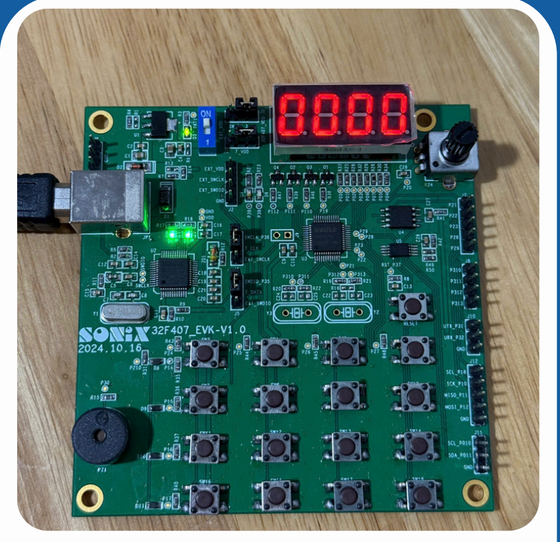
59 Minute



1 Hour



23 Hour 59 Minutes

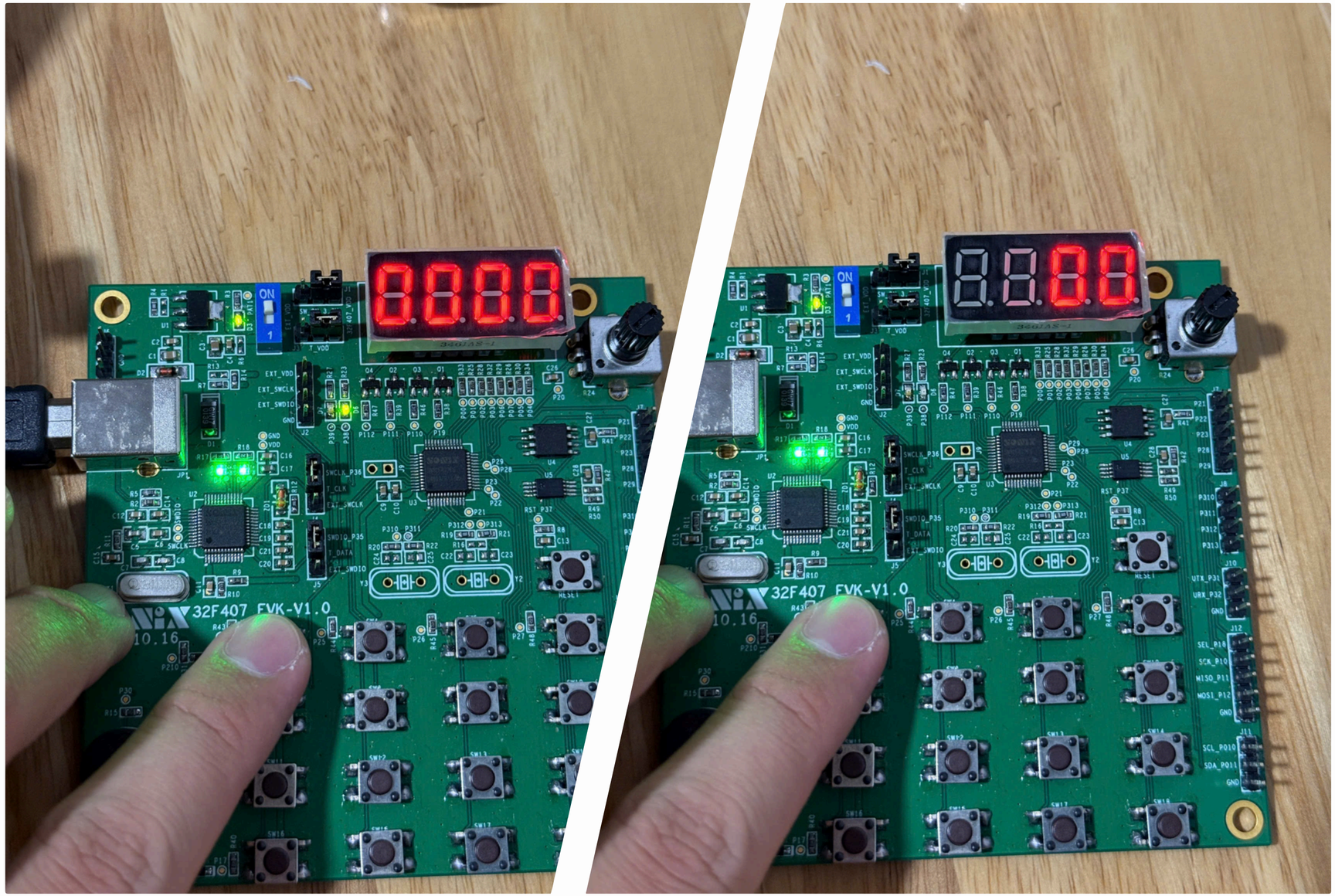


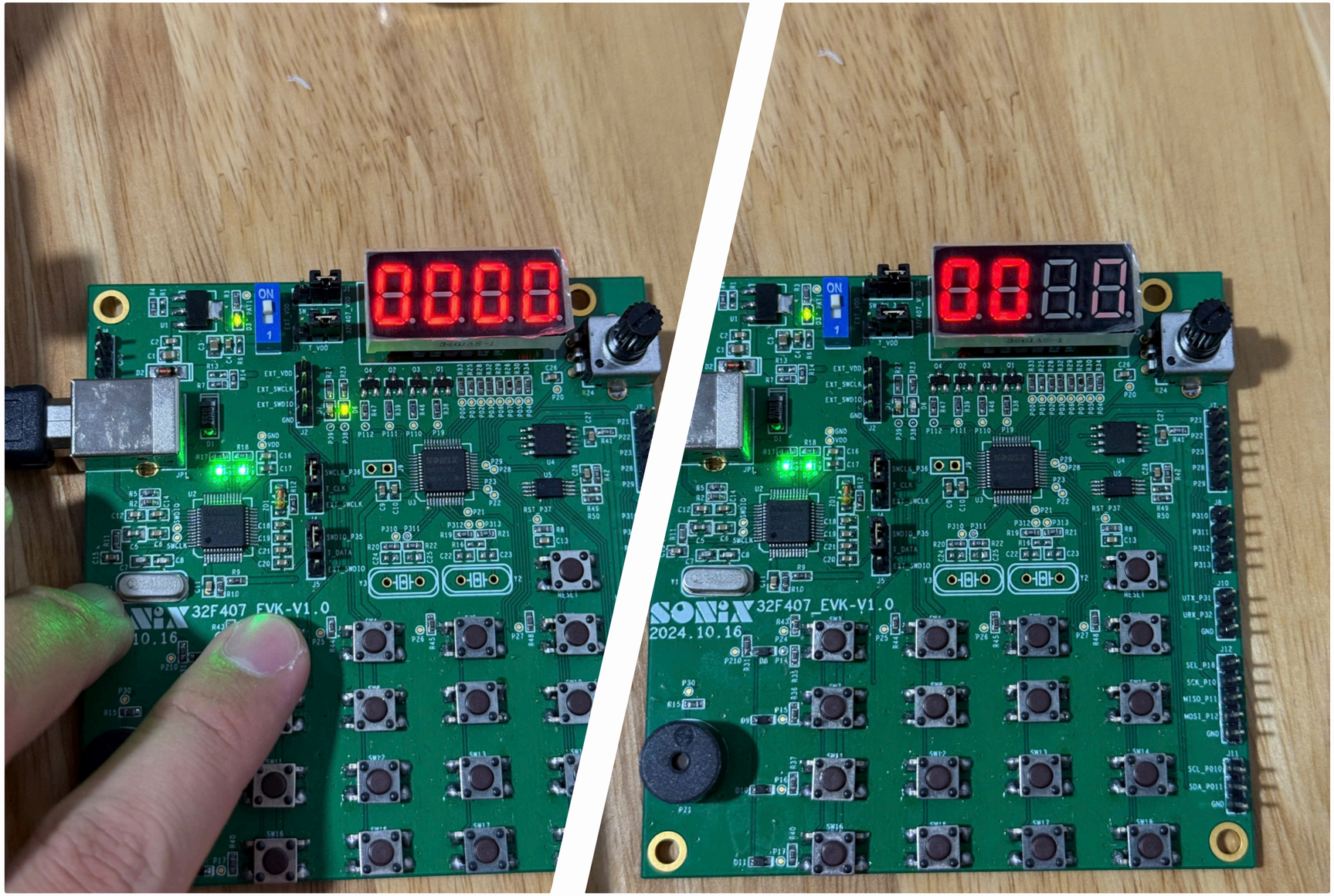
00 Hour 00 Minutes

Requirement 2:

Problem 01

·While in normal mode, press SW3 to enter time-setting mode: the HH 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).





Requirement 2:

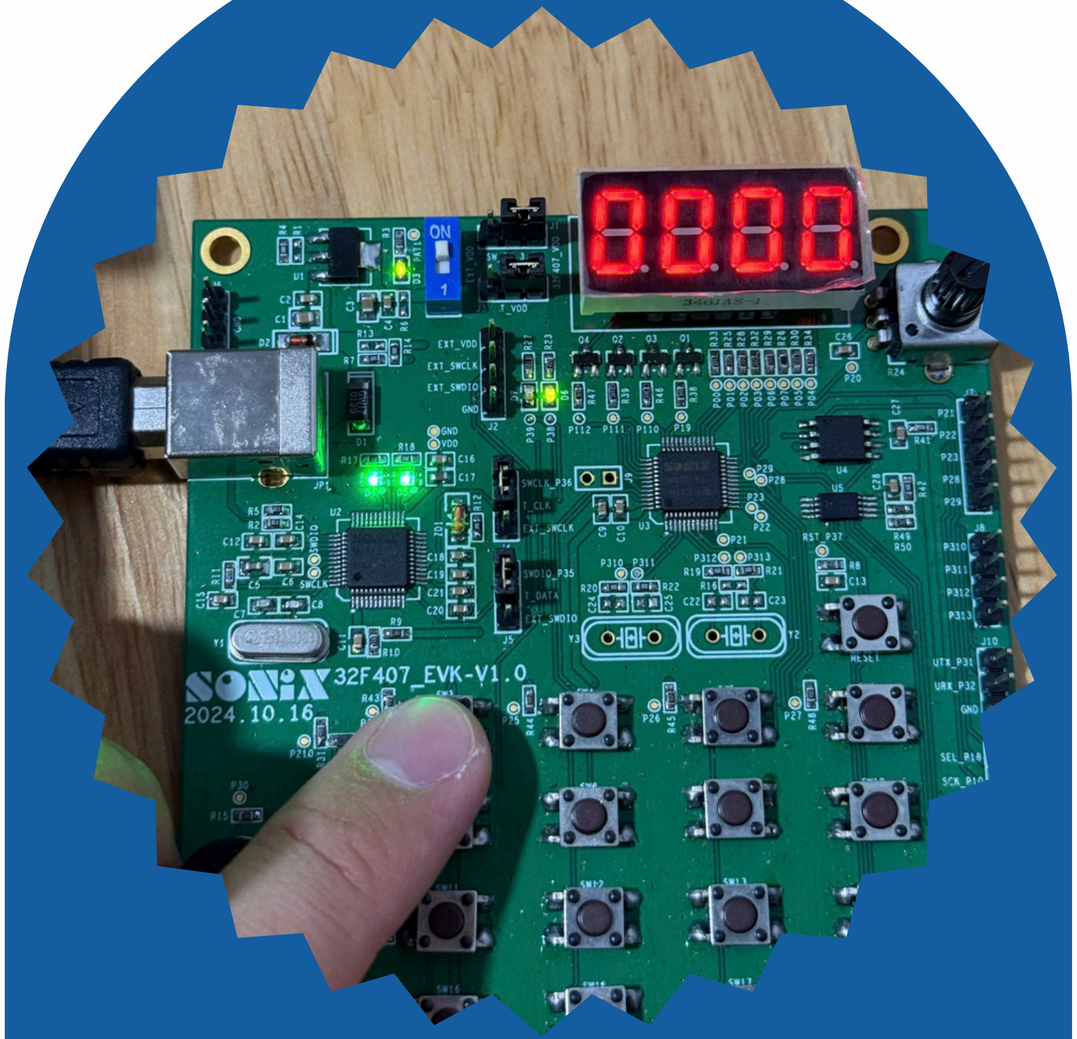
Problem 02

·While in hour-adjustment mode, press SW3 to enter minute-adjustment mode: the MM 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

Requirement 2:

Problem 03

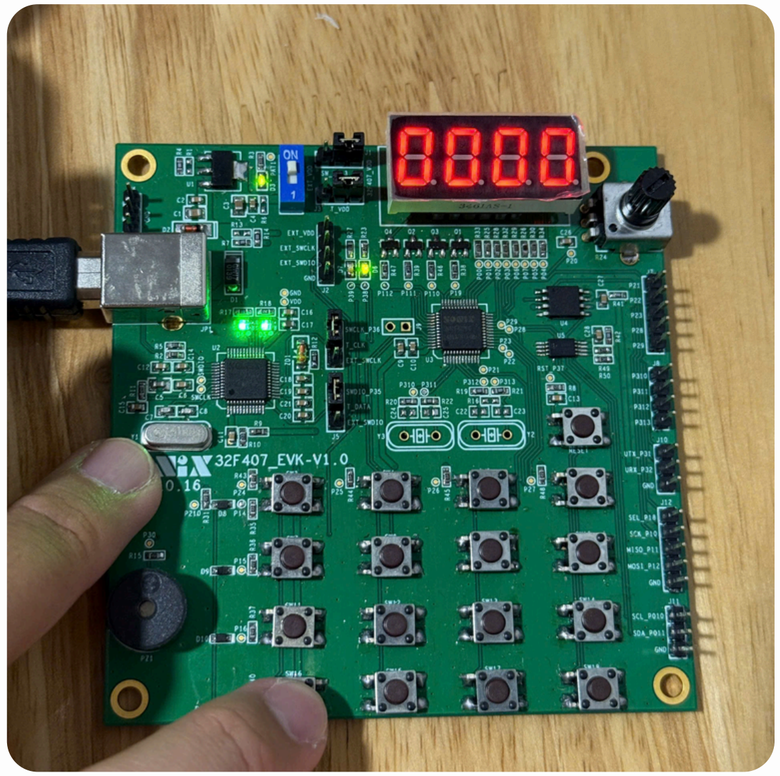
·While in minute-adjustment mode, press SW3 to enter normal mode.

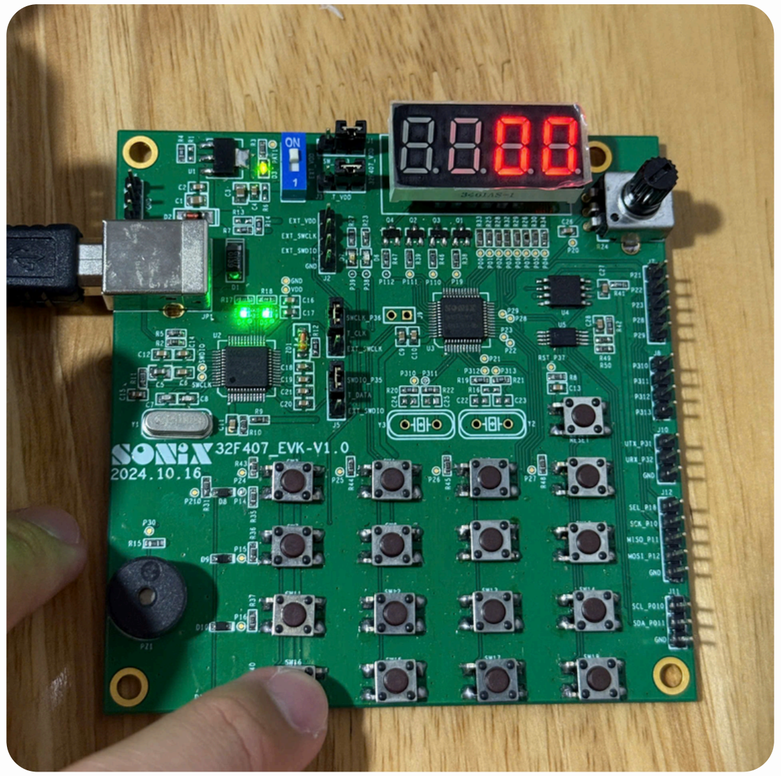


Requirement 3:

Problem 01

·While in normal mode, press SW16 to enter the scheduled time adjustment mode: the HH 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

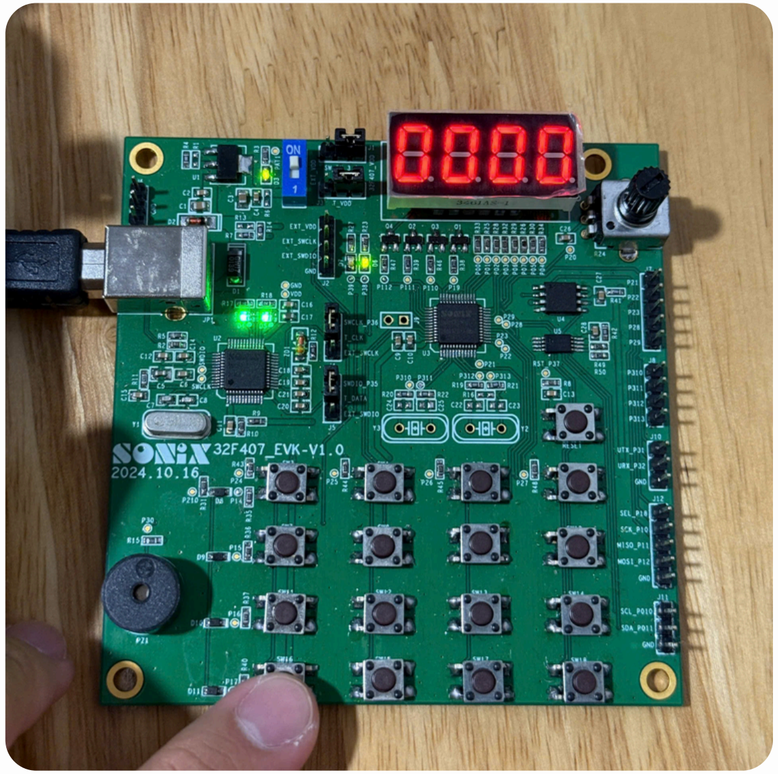


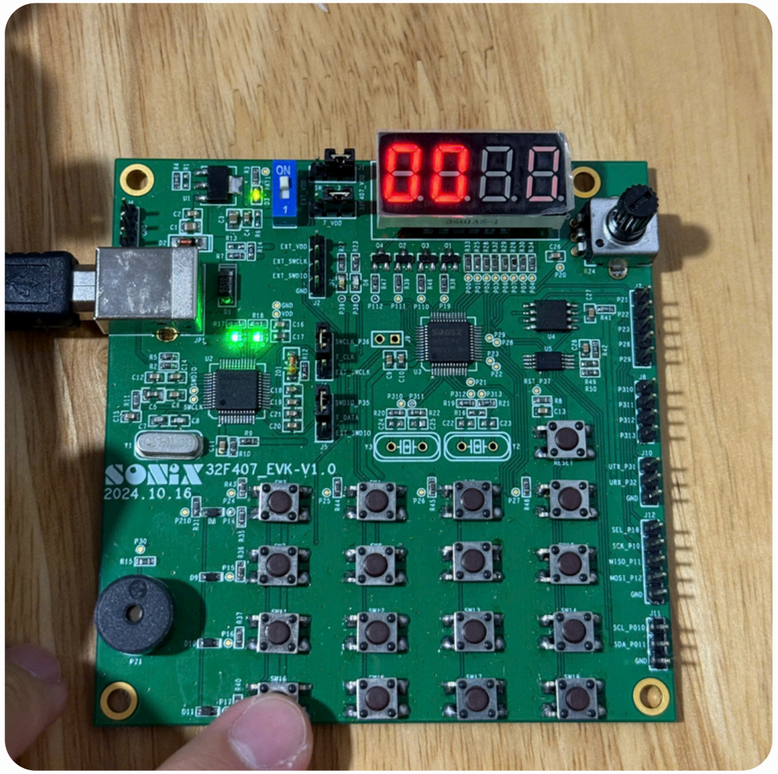


Requirement 3:

Problem 02

·While in the scheduled time adjustment mode, press SW16 to enter the scheduled minute adjustment mode: the MM 7-segment LED will flash with a 1-second cycle (0.5s ON – 0.5s OFF).

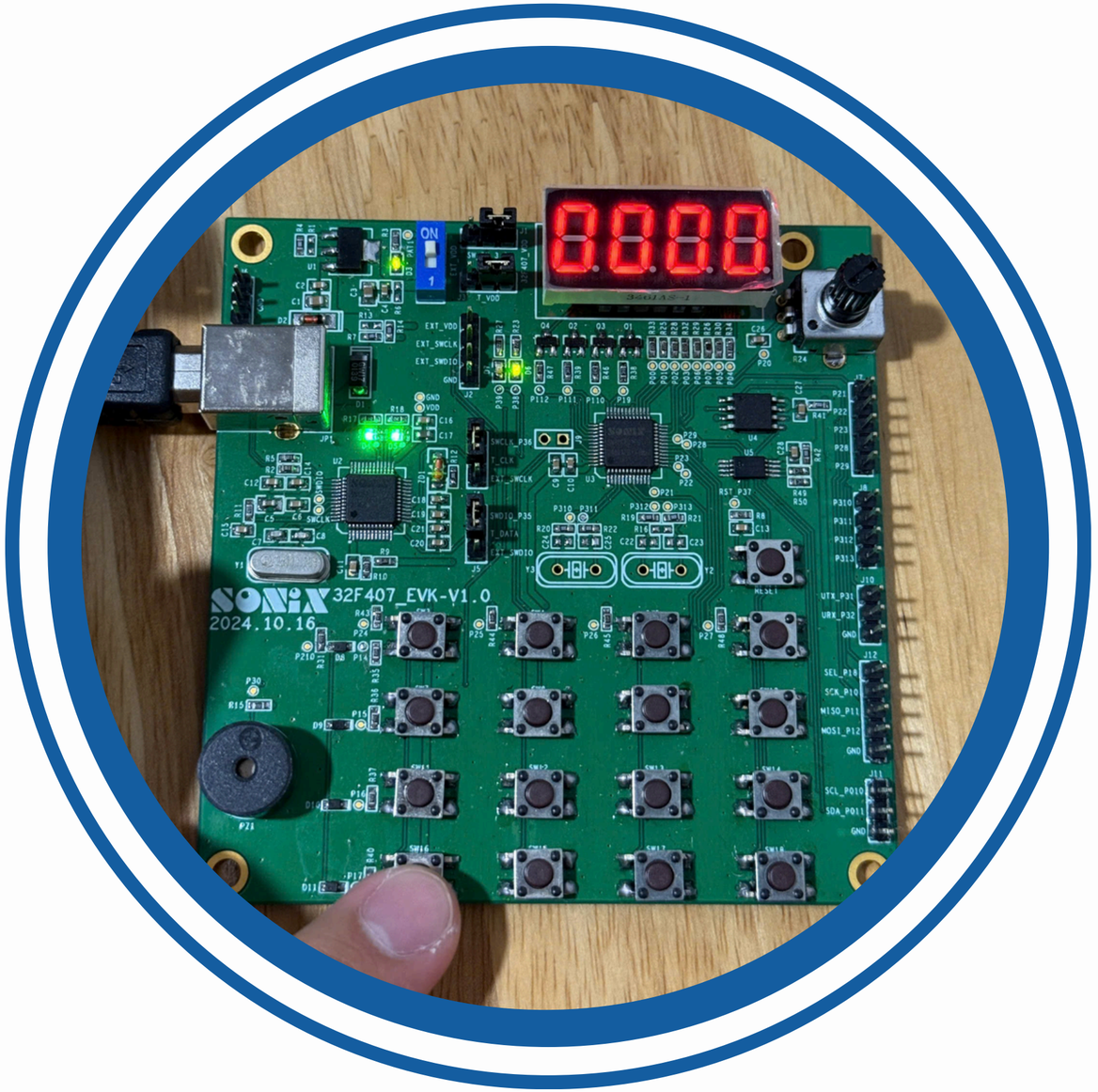




Requirement 3:

Problem 03

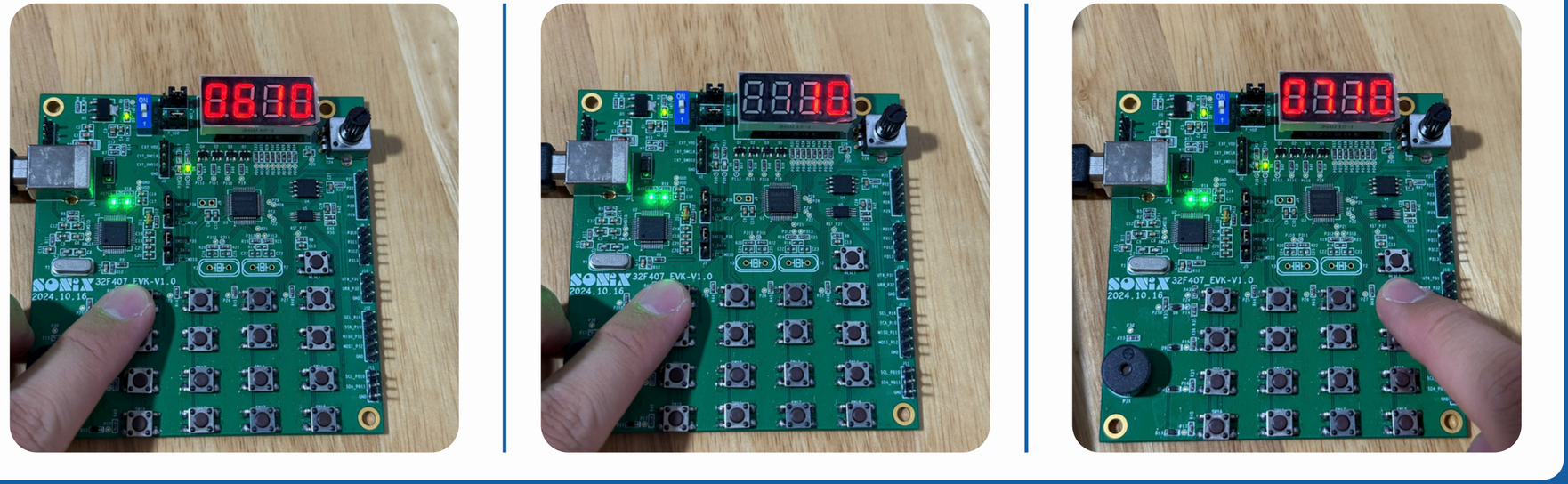
·While in the appointment minute adjustment mode, press SW16 to save the appointment time to EEPROM and exit to normal mode.



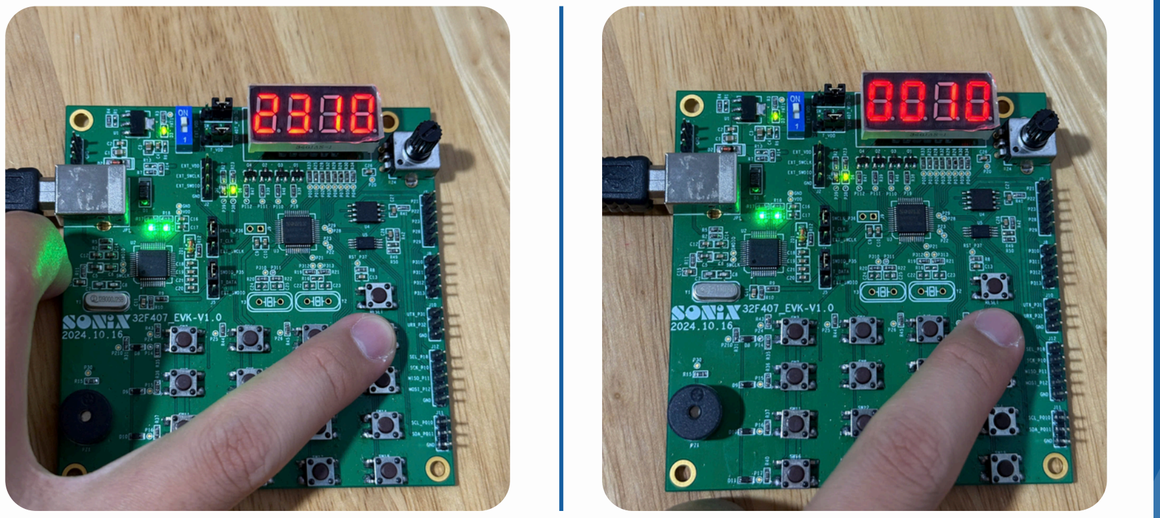
Requirement 4:

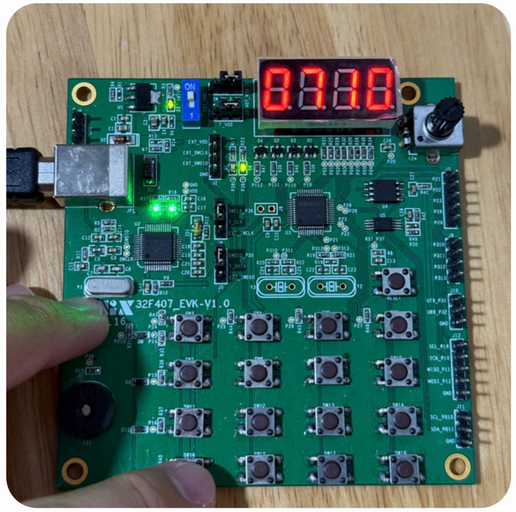
Problem 01

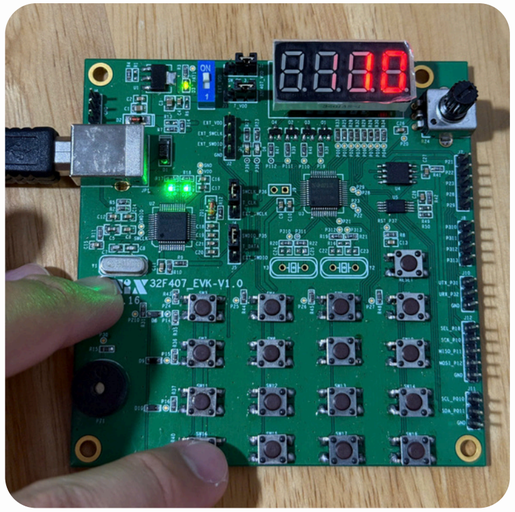
·While in time-setting or alarm-setting mode, press SW6 to increment the hour by 1; the hour resets to 0 upon reaching 24.

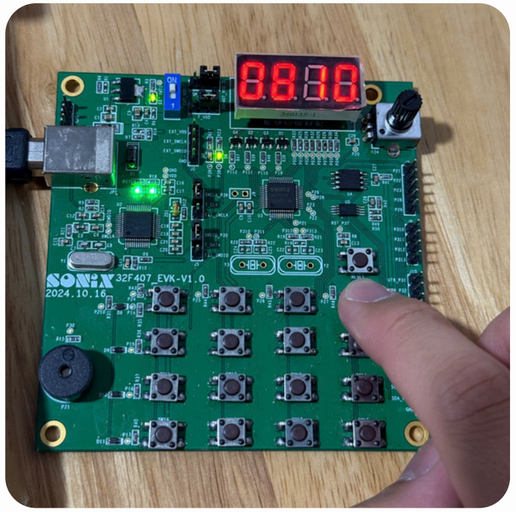


Increase Hours Setting

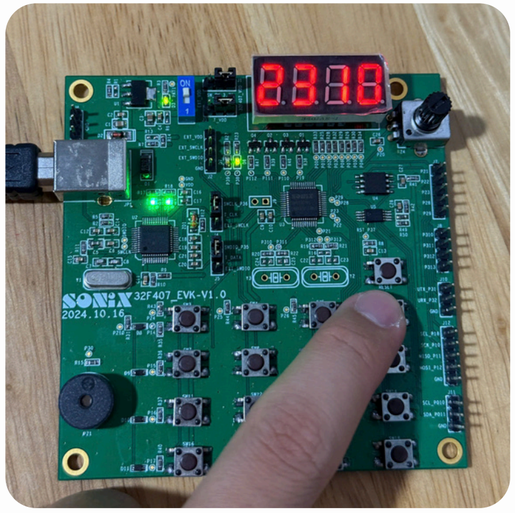


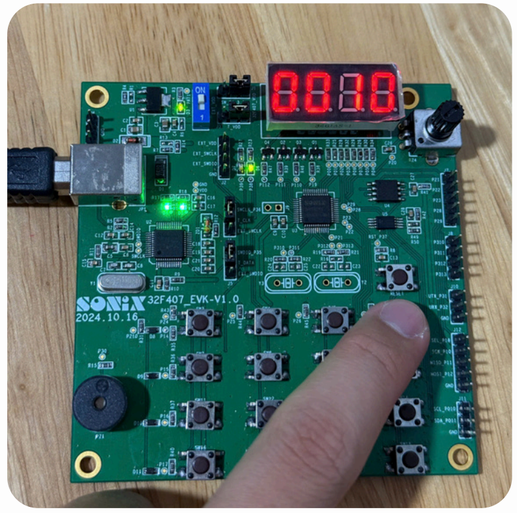






Increase Alarm Hours Setting

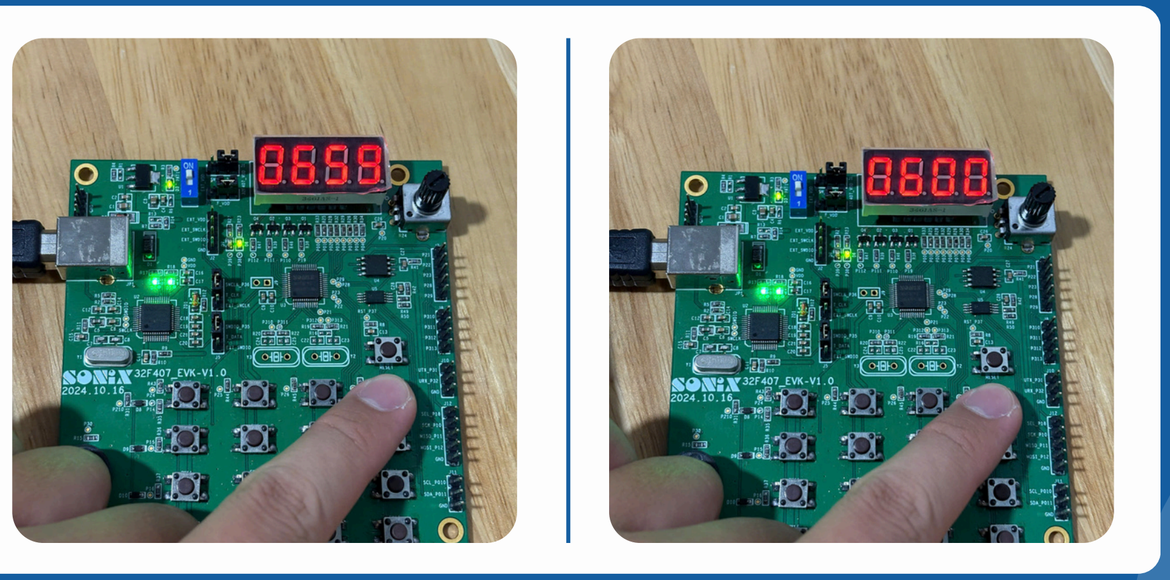




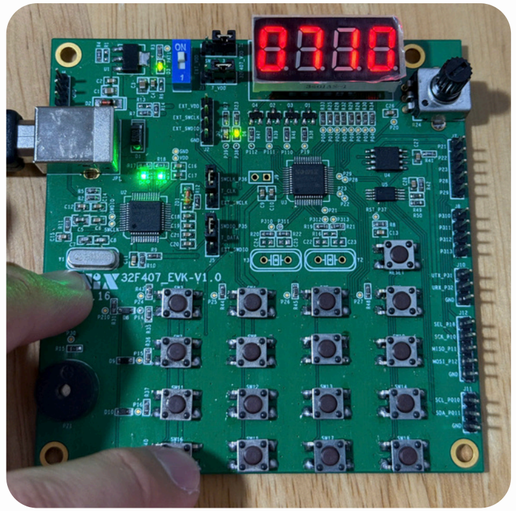
Requirement 4:

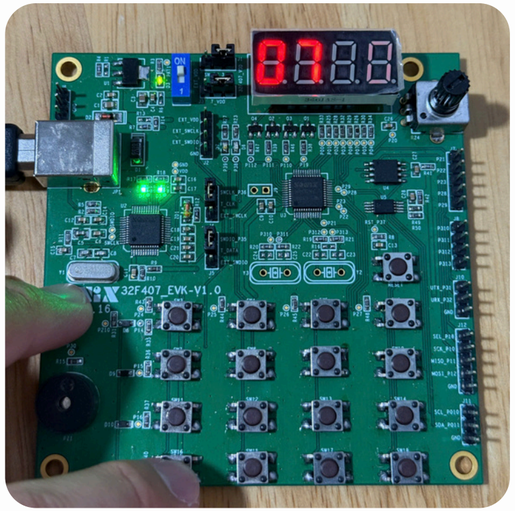
Problem 02

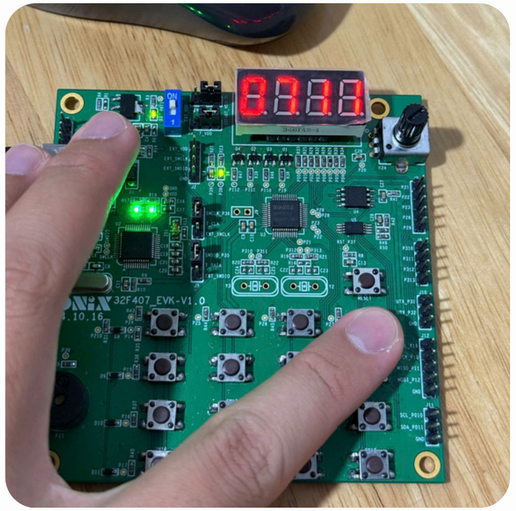
·While in minute adjustment mode or alarm minute adjustment mode, press SW6 to increment the minutes by 1; when the minutes reach 60, they will reset to 0.



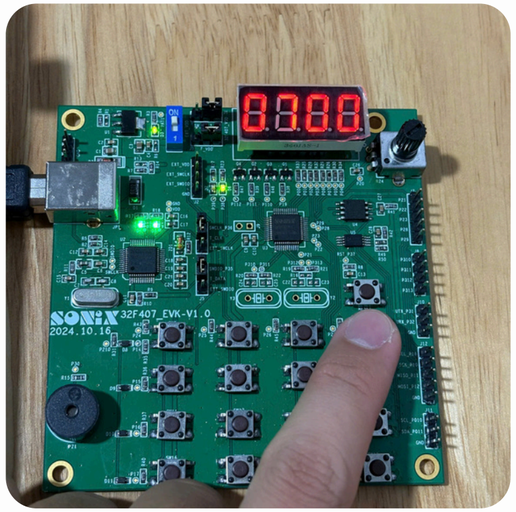
Increase Minutes Setting







Increase Alarm Minutes Setting

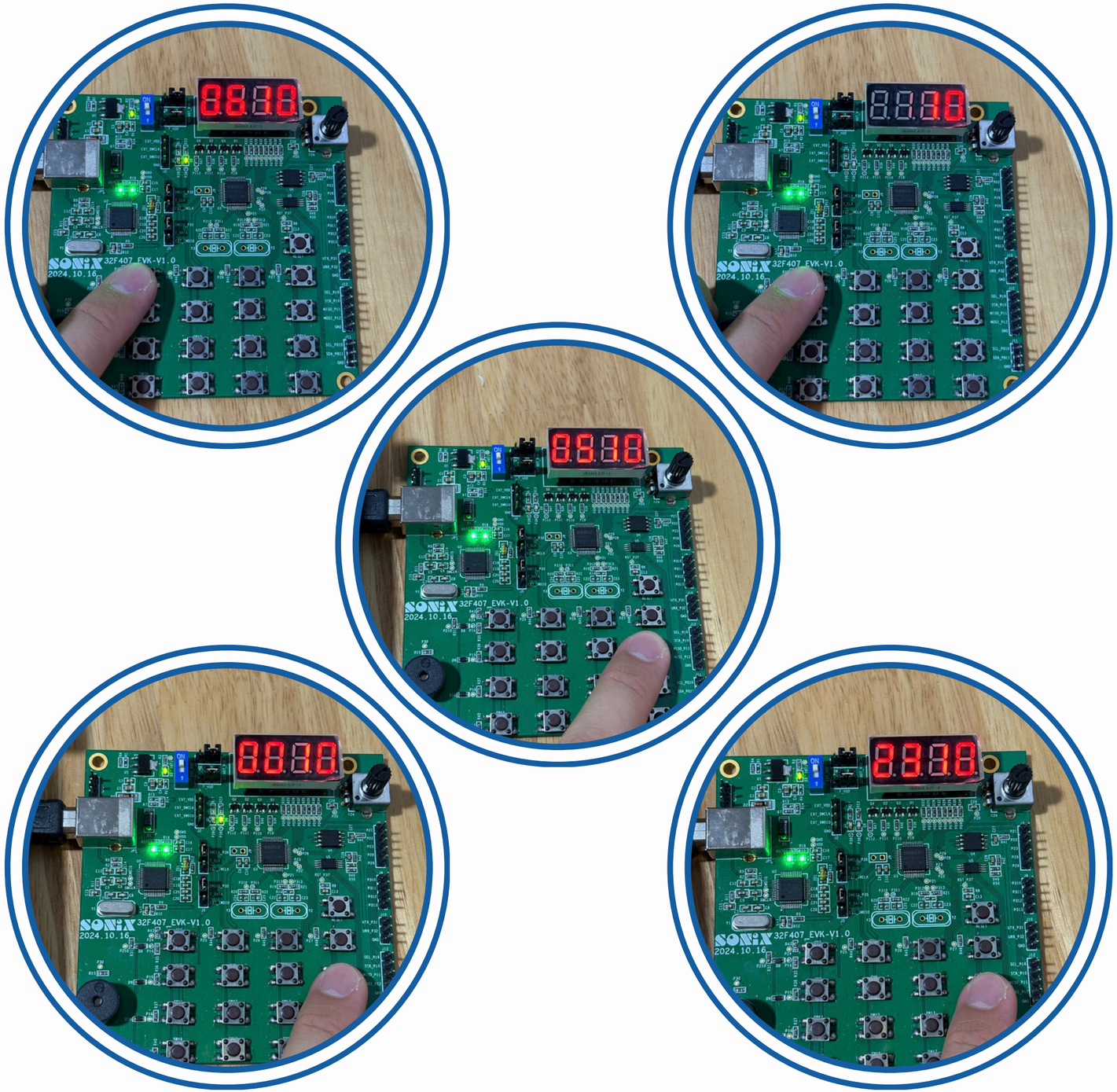


Requirement 5:

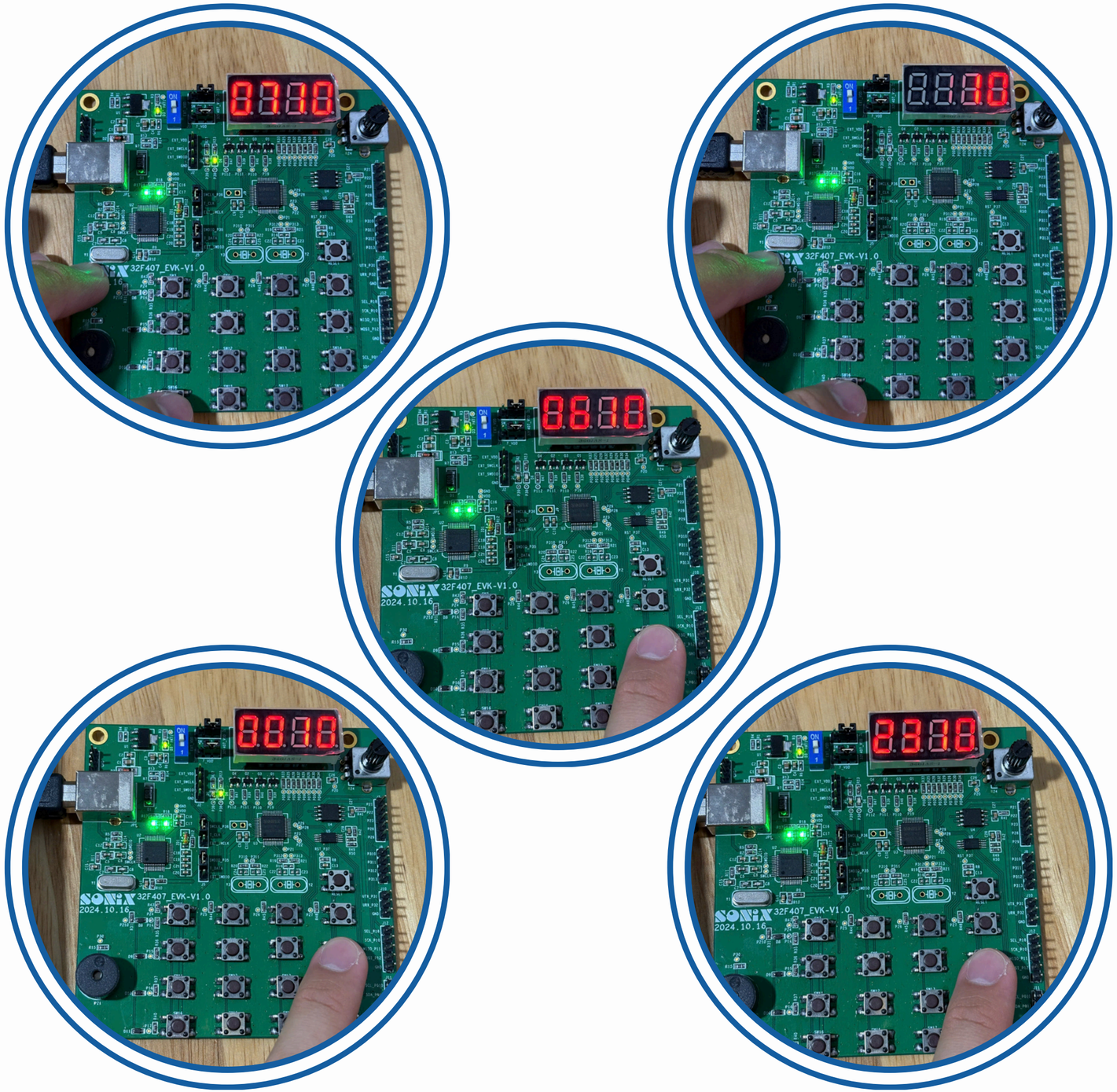
Problem 01

In time-setting or alarm-time-setting mode, press SW10 to decrease the hour by 1; if the hour is 0, decreasing it by 1 sets it to 23.

Decrease Hours Setting



Decrease Alarm Hours Setting

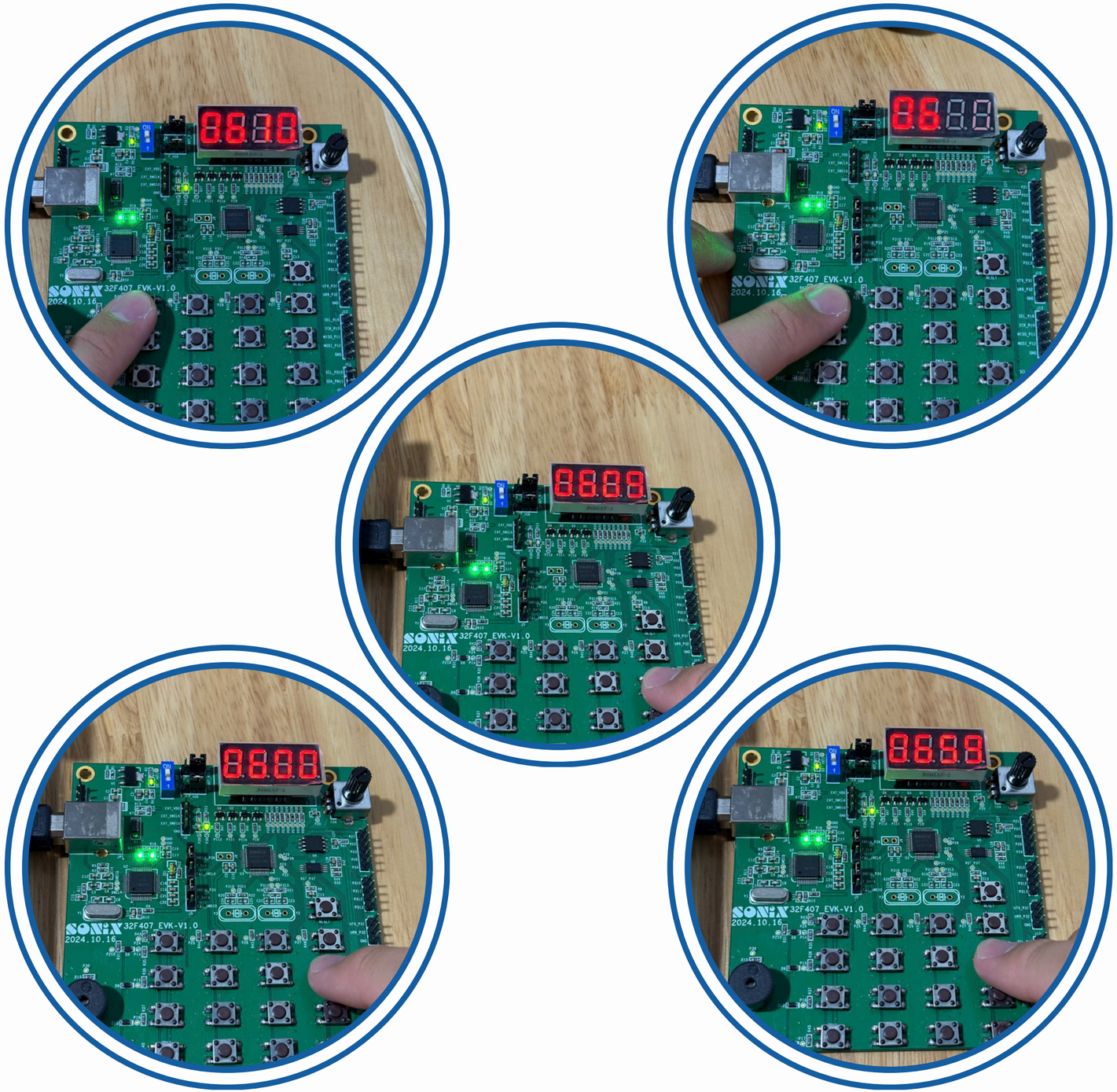


Requirement 5:

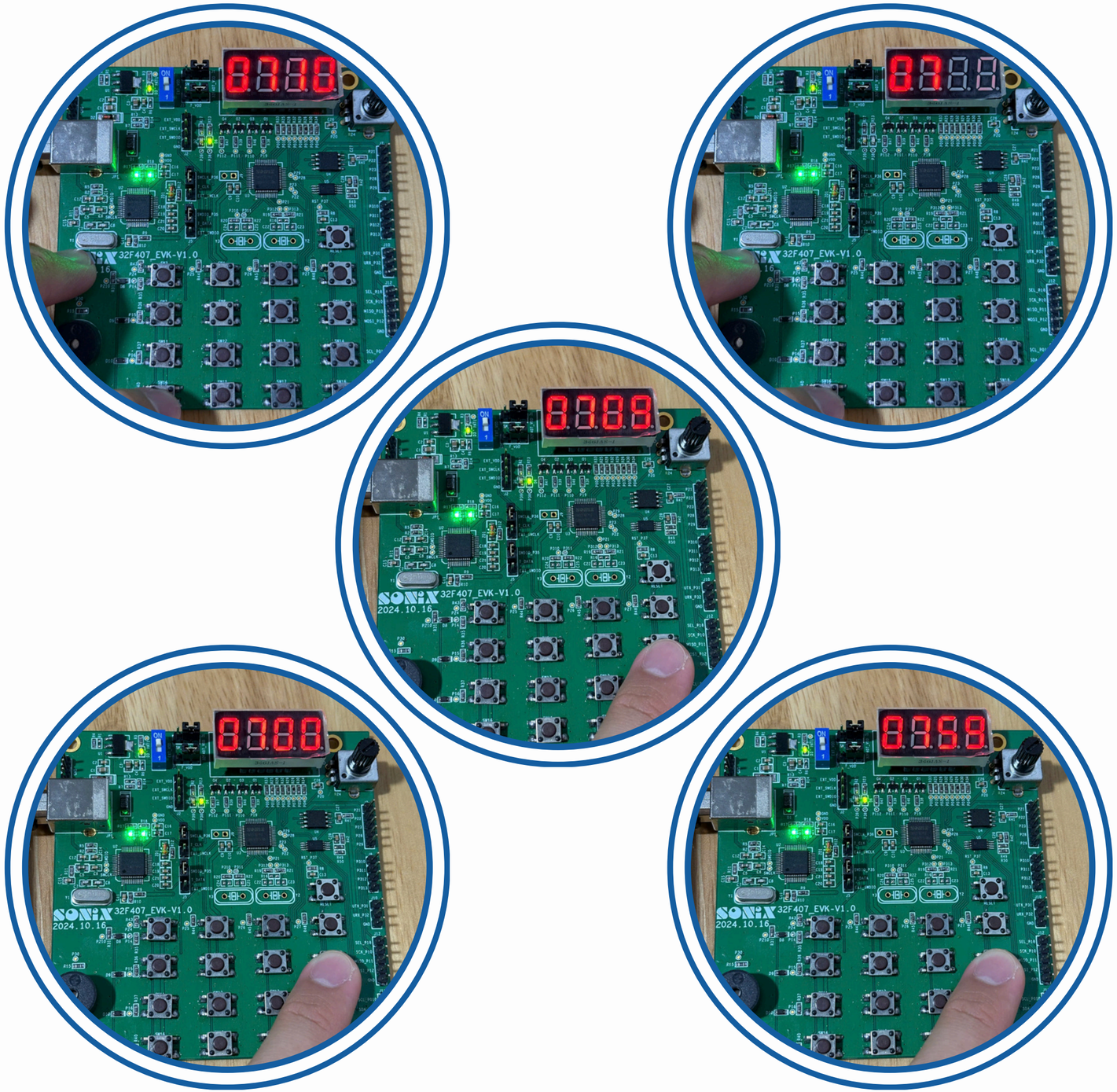
Problem 02

·In minute-setting or alarm-minute-setting mode, press SW10 to decrease the minute by 1; if the minute is 0, decreasing it by 1 sets it to 59.

Decrease Minutes Setting



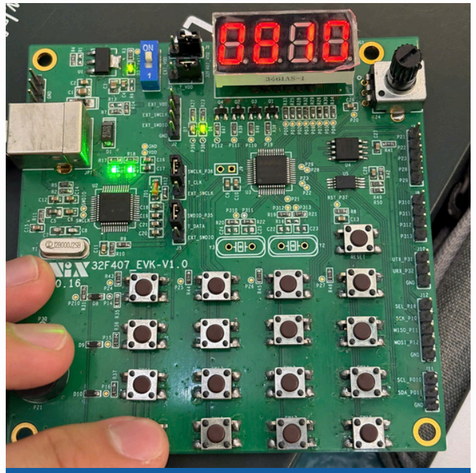
Decrease Alarm Minutes Setting

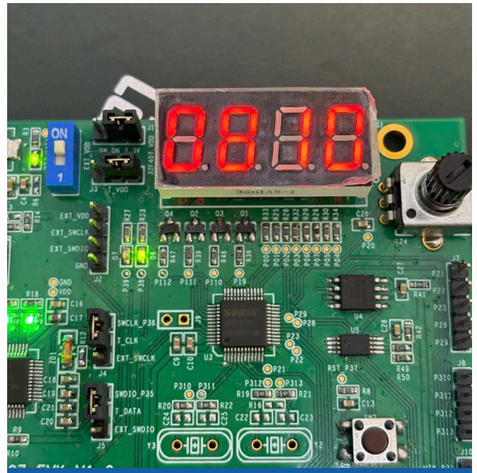


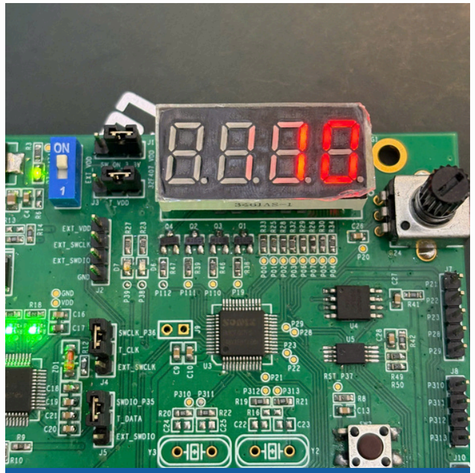
Problem 01

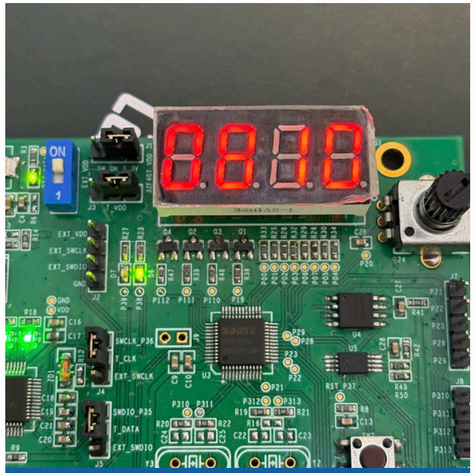
Requirement 7:

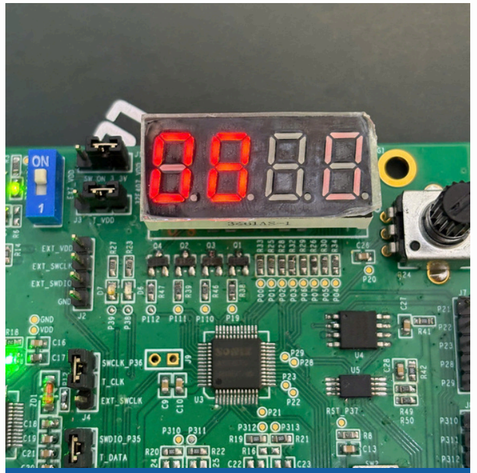
·In alarm hour or alarm minute adjustment modes, the LED flashes with a 1s cycle (0.5s ON – 0.5s OFF). Otherwise, the LED remains OFF.











Requirement 6, 8, 9:

- Pressing buttons SW3, SW6, SW10, or SW16 triggers a 0.3s beep.
- When the clock reaches the scheduled alarm time, the buzzer emits a continuous "beep-beep" pattern for 5 seconds (0.5s ON – 0.5s OFF cycle).
- Upon timing out and exiting the hour/minute adjustment mode or the alarm hour/minute adjustment mode to return to normal mode, the buzzer emits a 0.3s beep.

- ·Stores the scheduled alarm hour and minute; default alarm seconds = 0.

- ·In hour/minute adjustment mode or alarm hour/minute adjustment mode, if no button is pressed within 30 seconds, the system exits to normal mode.

(SEE THE RESULTS IN THE DEMO VIDEO)

THANK YOU!!

FOR WATCHING